$$h(m) = \delta(m) - \delta(m-4)$$

a) $x(m) = \cos(\omega_0 m T_s) \longrightarrow$ "$y(m) = A\cos(\omega_0 m T_s + \varphi)$"

$$y(m) = \sum_{k=-\infty}^{\infty} x(k) \cdot h(m-k) \Rightarrow y(m) = \sum_{k=-\infty}^{\infty} \cos(\omega_0 m T_s) \cdot [\delta(m-k) - \delta(m-k-4)]$$

$$\left.\begin{array}{l} x(m) * \delta(m) = x(m) \\ x(m) * \delta(m-4) = x(m-4) \end{array}\right\} y(m) = x(m) - x(m-4)$$

$$y(m) = \cos(\omega_0 m T_s) - \underbrace{\cos(\omega_0 (m-4) T_s)}$$

- $\cos(a-b) = \cos a \cos b + \operatorname{sen} a \operatorname{sen} b$

$$\cos(\omega_0 T_s (m-4)) = \cos(\omega_0 T_s m)\cos(4\omega_0 T_s) + \operatorname{sen}(\omega_0 T_s m) \cdot \operatorname{sen}(4\omega_0 T_s)$$

$$y(m) = \cos(\omega_0 m T_s) - [\cos(\omega_0 T_s m)\cos(4\omega_0 T_s) + \operatorname{sen}(\omega_0 T_s m) \cdot \operatorname{sen}(4\omega_0 T_s)]$$

$$y(m) = \underbrace{[1 - \cos(4\omega_0 T_s)]\cos(\omega_0 m T_s)}_{A} + \underbrace{\operatorname{sen}(\omega_0 T_s m) \cdot \operatorname{sen}(4\omega_0 T_s)}_{B}$$

- $A\cos(\omega_0 m T_s + \varphi) = \underbrace{A\cos(\omega_0 m T_s)\cos(\varphi)}_{A} - \underbrace{A\operatorname{sen}(\omega_0 m T_s)\operatorname{sen}(\varphi)}_{B}$

- Busco $A \longrightarrow A = \sqrt{[1 - \cos(4\omega_0 T_s)]^2 + [\operatorname{sen}(4\omega_0 T_s)]^2}$

$$A = \sqrt{1 - 2\cos(4\omega_0 T_s) + \underbrace{\cos^2(4\omega_0 T_s) + \operatorname{sen}^2(4\omega_0 T_s)}_{=1}}$$

- $1 - \cos(x) = 2\operatorname{sen}^2\left(\frac{x}{2}\right)$

$2 - 2\cos(x) = 4\operatorname{sen}^2\left(\frac{x}{2}\right)$

$$A = \sqrt{2 - 2\cos(4\omega_0 T_s)}$$

$$A = 2\,|\operatorname{sen}(2\omega_0 T_s)|$$

ENTONCES = $\underbrace{y(m) = 2\,|\operatorname{sen}(2\omega_0 T_s)|\cos(\omega_0 m T_s + \varphi)}$

b) $x(m) = (1/2)^m u(m)$

$x(m) = (1/2)^m u(m) \Rightarrow y$ [illegible]

$$y_{(n)} = \frac{1}{2^n} u_{(n)} - \frac{1}{2^{n-4}} u_{(n-4)} = \left(\frac{1}{2}\right)^n \left[u_{(n)} - 16\, u_{(n-4)}\right]$$

$n < 0 \rightarrow u_{(n)} = 0 \;/\; u_{(n-4)} = 0 \Rightarrow y_{(n)} = 0$

$n = 0 \rightarrow u_{(n)} = 1 \;/\; u_{(n-4)} = 0 \Rightarrow y_{(0)} = \delta_{(n)}$

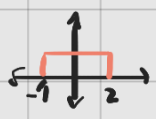
$\vdots$

$n = 4 \rightarrow u_{(n)} = 1 \;/\; u_{(n-4)} = 1 \Rightarrow y_{(4)} = \frac{1}{16}\left(\delta_{(n)} - 16\,\delta_{(n-4)}\right)$

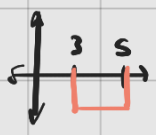
c) $x_{(n)} = u_{(n+1)} - u_{(n-2)}$

$y_{(n)} = x_{(n)} * \left(\delta_{(n)} - \delta_{(n-4)}\right)$

$y_{(n)} = x_{(n)} * \delta_{(n)} - x_{(n)} * \delta_{(n-4)}$



$x_{(n)} \begin{cases} 1 & -1 \leq n < 2 \\ 0 & \text{otro} \end{cases}$



$x_{(n-4)} \begin{cases} 1 & 3 \leq n \leq 5 \\ 0 & \text{otro} \end{cases}$

$$y_{(n)} = \underbrace{u_{(n+1)} - u_{(n-2)}}_{x_{(n)}} - \underbrace{u_{(n-3)} - u_{(n-6)}}_{x_{(n-4)}}$$